

Software Requirements Specification: Banking Management System

1. Project Overview

The Banking Management System is a comprehensive digital platform designed to facilitate secure financial transactions, account management, loan processing, and personalized financial analytics for customers, while providing administrative oversight and automated fraud mitigation for the institution.

2. Executive Summary

This system aims to modernize the banking experience by integrating core retail banking services with AI-driven financial health tools. The objective is to provide a seamless, secure, and compliant environment for customers to manage their finances while enabling internal staff to streamline loan approval workflows and administrative operations.

3. Stakeholders

- **Customers:** Individual users managing personal finances.
- **Bank Employees:** Loan officers and support staff handling manual approvals.
- **Administrators:** System overseers managing configurations and audit logs.
- **Compliance/Audit Team:** Users responsible for regulatory reporting and fraud investigation.

4. Target Users

- **Retail Customers:** Use the platform for daily banking, budgeting, and loan applications.
- **Bank Staff:** Use the platform to view customer data, process loans, and handle inquiries.
- **System Admins:** Manage user roles, system settings, and security parameters.

5. Functional Requirements

- **Account Management:** Users can open accounts and manage personal profile settings.
- **Transaction Management:** Support for internal transfers, peer-to-peer transfers, and bill payments.
- **Loan Management:** Customers can apply for loans; employees can review, approve, or deny them based on system criteria.
- **Security:** Implementation of Multi-Factor Authentication (MFA) for all sensitive actions.
- **Fraud Detection:** Automated flagging of suspicious transaction patterns.
- **AI Insights:** Automated spending analysis and generation of budget recommendations.
- **Auditing:** Comprehensive logging of all system transactions and administrative actions.

6. Non-Functional Requirements

- **Security:** Data must be encrypted at rest and in transit (AES-256/TLS 1.3).
- **Availability:** The system shall maintain 99.9% uptime.
- **Performance:** Transaction processing must complete within 2 seconds.
- **Compliance:** Must adhere to GDPR, PCI-DSS, and relevant banking regulations.
- **Scalability:** System must support concurrent scaling during peak banking hours.

7. User Stories

- *As a Customer*, I want to transfer money to another account so that I can pay expenses.
- *As a Customer*, I want to see a monthly breakdown of my spending so I can manage my budget.
- *As an Employee*, I want to view a list of pending loan applications so I can approve qualified borrowers.
- *As an Admin*, I want to view audit logs to ensure system security.
- *As a User*, I want to receive a prompt for MFA so that my account remains secure.

8. Acceptance Criteria

- Transaction must not be processed if MFA fails.

- Loan application status updates must trigger notifications to the customer.
- Fraud alerts must be generated within 5 seconds of a detected anomaly.
- Budgeting recommendations must refresh every 24 hours based on recent transaction data.

9. Business Rules

- Transfers exceeding a certain threshold (\$X) require secondary approval.
- Loan approval is prohibited if the credit score (via external API integration) is below Y.
- Accounts must remain active for 30 days before being eligible for loan applications.

10. Assumptions

- The system will integrate with a third-party credit bureau for credit score verification.
- The system will utilize an existing secure notification service (SMS/Email/Push).
- Users will have stable internet connectivity.

11. Constraints

- Must comply with local banking regulatory frameworks.
- Budgeting features are limited to internal bank data.
- All interactions must be logged in an immutable format.

12. Risks

- **Data Breach:** Unauthorized access to sensitive financial records.
- **Model Bias:** AI spending insights providing inaccurate recommendations.
- **Service Outage:** Third-party provider failure preventing authentication.

13. Out of Scope

- Physical branch ATM hardware integration.

- Stock trading or cryptocurrency brokerage services.
- International wire transfer (SWIFT) orchestration (Phase 2).
- Direct bank-to-bank core ledger hardware management.

14. Open Questions

- What is the specific maximum transaction threshold before manual oversight is required?
- Which specific external credit scoring services should be integrated?
- What is the acceptable latency for AI insight generation?
- Are there existing identity verification (KYC) providers already in use?